Product Requirements Document

Professional Mobile Camera System with Real-Time Lens & Image Manipulation

Version: 2.0
Date: October 11, 2025
Status: Complete Technical Specification

Executive Summary

A comprehensive mobile photography application that eliminates the need for professional camera equipment by providing real-time lens simulation, advanced image manipulation, manual camera controls, and composition guidance—all in a single, intuitive interface. This app transforms the smartphone into a complete professional photography system with computational photography capabilities that rival and, in some aspects, exceed traditional DSLR/mirrorless camera workflows.

Vision Statement

"The last camera you'll need to carry."

This application provides photographers with complete creative control over every aspect of image capture and manipulation in real-time, combining professional manual controls, computational lens effects, advanced color grading, and composition guidance into a unified, intuitive experience.

Core Problem Statement

Professional and enthusiast photographers currently face a multi-device, multi-app workflow:

1. **Equipment Burden:** DSLR/mirrorless camera + multiple lenses + phone
2. **Fragmented Workflow:** Shoot → Transfer → Edit in separate software
3. **Limited Real-Time Feedback:** Can't see final result until post-processing
4. **No Composition Learning:** Built-in guides are basic or non-existent
5. **Complexity:** Different interfaces for camera settings, lens effects, and editing

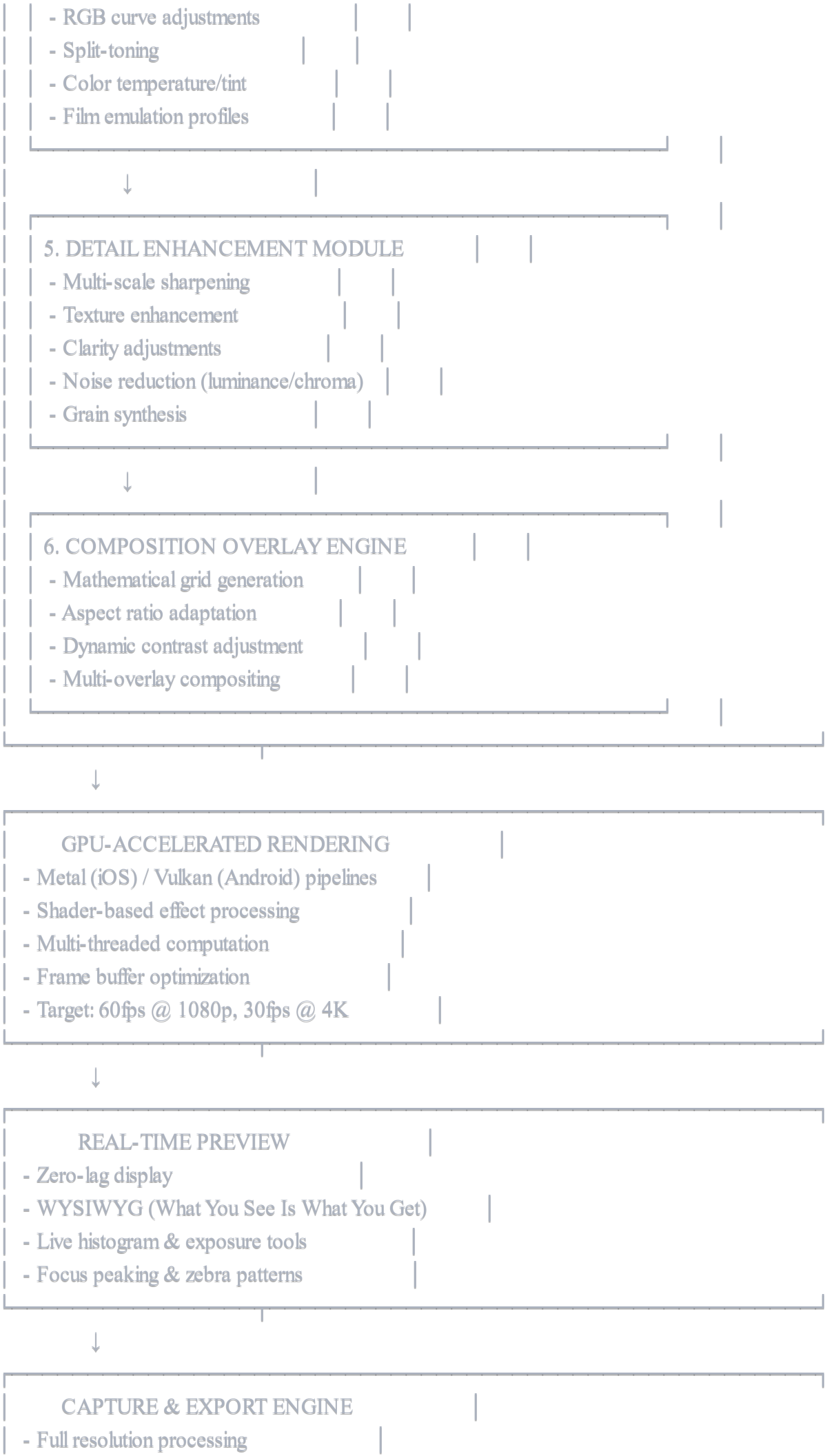
Solution: One mobile application that handles everything from lens simulation and capture to color grading and composition—all with real-time preview showing the exact final result.

Technical Foundation: The Core Processing Pipeline

Real-Time Image Processing Architecture




CAMERA INPUT LAYER
- Multi-lens access (Ultrawide, Wide, Telephoto)
- RAW sensor data capture
- Depth sensor integration (dual camera/LiDAR)
- Hardware-accelerated preview stream
COMPUTATIONAL PHOTOGRAPHY ENGINE
1. DEPTH PROCESSING MODULE
- Real-time depth map generation
- Multi-frame depth refinement
- Edge-aware depth smoothing
- Depth confidence mapping
2. LENS SIMULATION MODULE
- Aperture simulation (f/1.2 - f/22)
- Bokeh rendering engine
- Focal length emulation
- Lens distortion/correction
- Chromatic aberration control
- Vignetting simulation
3. EXPOSURE & SENSOR SIMULATION
- ISO noise modeling
- Shutter speed effects
- Motion blur synthesis
- HDR merge & tone mapping
- Highlight recovery
- Shadow detail extraction
4. COLOR GRADING ENGINE
- 3D LUT processing
- HSL manipulation per channel


- RGB curve adjustments
- Split-toning
- Color temperature/tint
- Film emulation profiles
5. DETAIL ENHANCEMENT MODULE
- Multi-scale sharpening
- Texture enhancement
- Clarity adjustments
- Noise reduction (luminance/chroma)
- Grain synthesis
6. COMPOSITION OVERLAY ENGINE
- Mathematical grid generation
- Aspect ratio adaptation
- Dynamic contrast adjustment
- Multi-overlay compositing
GPU-ACCELERATED RENDERING
- Metal (iOS) / Vulkan (Android) pipelines
- Shader-based effect processing
- Multi-threaded computation
- Frame buffer optimization
- Target: 60fps @ 1080p, 30fps @ 4K
REAL-TIME PREVIEW
- Zero-lag display
- WYSIWYG (What You See Is What You Get)
- Live histogram & exposure tools
- Focus peaking & zebra patterns
CAPTURE & EXPORT ENGINE
- Full resolution processing

- RAW + processed JPEG pairs
- Non-destructive editing data
- Metadata preservation

Feature Requirements: Complete Specification

1. LENS SIMULATION SYSTEM

1.1 Aperture Control & Depth of Field Simulation

Priority: P0 (Core Feature)

Capability: Simulate variable aperture from f/1.2 (extreme shallow depth) to f/22 (deep focus) on any scene, regardless of phone's physical lens limitations.

Technical Implementation:

- **Depth Map Generation:**
  - Utilize dual/triple camera systems for hardware depth sensing
  - Implement AI-based monocular depth estimation for single-camera devices
  - Real-time depth refinement using temporal coherence across frames
  - Edge-aware depth smoothing to prevent blur artifacts
- **Bokeh Rendering Engine:**
  - Physics-based circle of confusion calculation
  - Customizable bokeh shapes (circular, hexagonal, octagonal)
  - Simulated lens aberrations (spherical, chromatic)
  - Depth-dependent blur intensity
  - Neural rendering for photorealistic bokeh quality
- **Performance Optimization:**
  - GPU-accelerated blur kernels
  - Adaptive resolution processing (full res for still, reduced for preview)
  - Depth map caching and prediction

User Controls:

- **Aperture Slider:** f/1.2, 1.4, 1.8, 2.0, 2.8, 4.0, 5.6, 8.0, 11, 16, 22
- **Bokeh Shape:** Circular (default), Hexagonal, Octagonal, Custom
- **Bokeh Quality:** Preview (fast), High (balanced), Ultra (slow, capture only)
- **Focus Plane Visualization:** Real-time depth preview overlay
- **Touch to Focus:** Tap to set focus point and depth plane

Integration Notes:

- Works with all camera lenses (wide, ultrawide, telephoto)
- Automatically adjusts to scene depth complexity
- Disables gracefully on devices without depth capabilities
- Can use single-frame or multi-frame depth data

1.2 Focal Length Emulation

Priority: P0 (Core Feature)

Capability: Simulate different focal lengths beyond phone's physical lenses (14mm ultrawide to 200mm+ telephoto) with intelligent quality preservation.

Technical Implementation:

- **Digital Zoom Enhancement:**
  - Multi-frame super-resolution for zoomed captures
  - AI-based detail synthesis for digital zoom beyond optical range
  - Intelligent cropping with upscaling algorithms
- **Perspective Correction:**
  - Automatic lens distortion correction for ultrawide
  - Perspective shift for architectural photography
  - Focal length-specific compression simulation
- **Lens Character Simulation:**
  - Wide angle: edge distortion, dynamic range challenges
  - Standard: neutral rendering
  - Telephoto: background compression, reduced depth of field range

User Controls:

- **Focal Length Slider:** Continuous from 14mm-200mm equivalent
- **Quick Presets:** 14mm, 24mm, 35mm, 50mm, 85mm, 135mm, 200mm
- **Crop vs. Quality Mode:** Prioritize framing or image quality
- **Lens Characteristics:** Enable/disable distortion, compression effects

Technical Specifications:

- Seamless switching between physical lenses as user adjusts
- Quality indicators showing optical vs. digital zoom
- Resolution preview showing capture size at selected focal length

1.3 Lens Effects & Corrections

Priority: P1 (Enhanced Feature)

Capabilities:

- **Distortion Control:**
  - Barrel distortion correction (ultrawide lenses)
  - Pincushion distortion correction (telephoto lenses)
  - Manual distortion adjustment slider
- **Vignetting:**
  - Natural lens vignetting simulation
  - Artistic vignetting with adjustable intensity
  - Edge darkening/lightening control
- **Chromatic Aberration:**
  - Automatic CA removal
  - Artistic CA addition for vintage effects
  - RGB channel shift controls
- **Lens Flare & Light Effects:**

- Simulated lens flare (adjustable position/intensity)
- Star burst effects at point light sources
- Light bloom and halation

User Controls:

- Individual toggle/sliders for each effect
- Preset combinations (Clean, Natural, Vintage, Cinematic)
- Real-time preview of all corrections
- Before/after comparison view

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2. MANUAL CAMERA CONTROLS

2.1 Exposure Triangle Control

Priority: P0 (Core Feature)

Capability: Full manual control over ISO, Shutter Speed, and Aperture (simulated) with professional-grade precision.

ISO Control:

- **Range:** Native minimum to maximum (device-dependent, typically 25-6400+)
- **Extended Range:** Computational ISO boost up to 25,600 equivalent
- **Implementation:**
  - Hardware ISO for minimum to base range
  - Computational noise modeling for extended range
  - Real-time noise preview
  - Auto ISO with configurable minimum/maximum

User Interface:

- Fine adjustment dial (±1/3 stop increments)
- Quick presets: 100, 200, 400, 800, 1600, 3200, 6400
- ISO priority mode (auto shutter/aperture)
- Exposure preview with clipping warnings

Shutter Speed Control:

- **Range:** 1/8000s to 30 seconds (hardware dependent)
- **Extended Range:** Computational long exposure up to 300 seconds
- **Implementation:**
  - Hardware shutter control where available
  - Multi-frame averaging for long exposures
  - Motion blur synthesis for fast shutter effects

Special Modes:

- Bulb mode (hold shutter indefinitely)
- Interval timer for time-lapse
- Motion blur amount preview
- Reciprocity calculator (for ND filter simulation)

User Interface:

- Shutter speed dial (1/3 stop increments)
- Quick access: 1/8000, 1/4000, 1/1000, 1/250, 1/60, 1/30, 1s, 2s, 5s, 30s
- Shutter priority mode (auto ISO/aperture)
- Motion indicator showing blur amount

Aperture Control:

- See Section 1.1 (Lens Simulation)
- Integrated with exposure triangle
- Real-time depth of field preview
- Aperture priority mode (auto ISO/shutter)

Exposure Compensation:

- Range: -5 to +5 EV
- 1/3 stop increments
- Quick access dial
- Real-time histogram update

2.2 Focus Control

Priority: P0 (Core Feature)

Focus Modes:

- **Manual Focus:**
  - Continuous focus distance slider
  - Focus peaking overlay (highlights in-focus areas)
  - Magnified focus assist (tap to zoom)
  - Focus distance scale (feet/meters)
  - Infinity lock
- **Auto Focus:**
  - Tap to focus
  - Continuous AF (tracking)
  - Face detection AF
  - Eye detection AF (human and animal)
  - AF area modes: single point, zone, wide
- **Focus Stacking:**
  - Automatic multi-shot focus stacking
  - Manual focus range selection
  - Live preview of stacking result
  - Export as DNG stack for post-processing

Focus Assist Tools:

- **Focus Peaking:** Colored highlight overlay (red, yellow, white, off)
- **Focus Magnifier:** 2x, 5x, 10x zoom on focus point
- **Focus Confirmation:** Audio/haptic feedback when focused
- **Hyperfocal Distance Indicator:** Shows optimal focus for max DOF

2.3 White Balance & Color Temperature

Priority: P0 (Core Feature)

White Balance Modes:

- **Auto White Balance:** AI-based scene detection
- **Presets:**
  - Daylight (5500K)
  - Shade (7500K)
  - Cloudy (6500K)
  - Tungsten (3200K)
  - Fluorescent (4000K)
  - Flash (5500K)
- **Custom Kelvin:** 2500K to 10,000K in 100K increments
- **Manual Gray Card:** Tap to set from neutral subject
- **Fine Tune Tint:** Green ↔ Magenta shift (-50 to +50)

User Interface:

- Kelvin temperature slider with live preview
- Preset buttons with one-tap access
- Tint adjustment slider
- Real-time color cast visualization
- Before/after comparison

2.4 Metering & Exposure Tools

Priority: P0 (Core Feature)

Metering Modes:

- **Evaluative/Matrix:** Full frame metering (default)
- **Center-Weighted:** Prioritizes center of frame
- **Spot Metering:** Single point (1-3% of frame)
- **Partial Metering:** ~10% center circle
- **Highlight Priority:** Preserves highlight detail
- **Shadow Priority:** Preserves shadow detail

Real-Time Exposure Feedback:

- **Live Histogram:**
  - RGB composite histogram
  - Individual R, G, B channel histograms
  - Luminance histogram
  - Real-time update as settings change
- **Zebra Patterns:**
  - Overexposure warning (configurable threshold 95-105%)
  - Underexposure warning (configurable threshold 0-10%)
  - Color-coded intensity (yellow warning, red clipping)
- **False Color Display:**
  - IRE/stop visualization overlay
  - Color-coded exposure zones
  - Skin tone indicator lines
- **Waveform Monitor:**
  - Luminance waveform display
  - RGB parade waveform
  - Vectorscope for color distribution

Advanced Exposure Features:

- **Exposure Lock (AE-L):** Lock current exposure values
- **Focus/Exposure Split:** Separate focus and metering points
- **Exposure Bracketing:**
  - 3, 5, 7, or 9 shot brackets
  - Configurable EV spacing (1/3, 1/2, 1, 2 stops)
  - Auto HDR merge option

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3. REAL-TIME IMAGE PROCESSING & COLOR GRADING

3.1 Tone & Exposure Adjustments

Priority: P0 (Core Feature)

Global Adjustments (Real-Time):

- **Exposure:** -5.0 to +5.0 EV
- **Contrast:** -100 to +100
- **Highlights:** -100 to +100 (recovery/suppress)
- **Shadows:** -100 to +100 (lift/crush)
- **Whites:** -100 to +100 (white point adjustment)
- **Blacks:** -100 to +100 (black point adjustment)

Technical Implementation:

- GPU-accelerated tone mapping
- 16-bit internal processing pipeline
- Histogram-based adjustment calculations
- Real-time preview with minimal latency (<16ms)
- Non-destructive adjustment stacking

User Interface:

- Individual sliders for each parameter
- Linked adjustment mode (auto-balance)
- Numeric value display
- Reset button per slider
- Before/after toggle
- Adjustment presets (save/load)

3.2 Tone Curves

Priority: P1 (Enhanced Feature)

Curve Types:

- **Master Curve:** Affects all channels (luminance)
- **RGB Curves:** Individual Red, Green, Blue channel curves
- **Parametric Curves:** Highlight, light, dark, shadow point control

Technical Implementation:

- Bezier curve interpolation
- 256-point lookup table generation
- Hardware-accelerated LUT application
- Curve presets (S-curve, flat, lifted blacks, crushed blacks)

User Controls:

- Touch-drag curve editor
- Multi-point curve control
- Curve preset library
- Curve copy/paste between channels
- Curve strength/opacity control

3.3 Color Grading Engine

Priority: P0 (Core Feature)

HSL Controls (Per Color Channel):

- **Hue Shift:** -180° to +180° for each color
  - Red, Orange, Yellow, Green, Aqua, Blue, Purple, Magenta
- **Saturation:** -100 to +100 per color
- **Luminance:** -100 to +100 per color

Global Color Controls:

- **Vibrance:** -100 to +100 (intelligent saturation boost)
- **Saturation:** -100 to +100 (global saturation)
- **Temperature:** 2500K to 10,000K
- **Tint:** -150 (green) to +150 (magenta)

Split Toning:

- **Highlight Tint:** Hue selector (0-360°) + saturation (0-100%)
- **Shadow Tint:** Hue selector (0-360°) + saturation (0-100%)
- **Balance:** -100 (shadows) to +100 (highlights)

Technical Implementation:

- HSV color space conversion
- GPU-accelerated color transformation matrices
- Real-time color picker with scene sampling
- Color space preservation (wide color gamut support)

3.4 Look-Up Tables (LUTs)

Priority: P1 (Enhanced Feature)

LUT System:

- **3D LUT Support:** Standard .cube format (17^3, 33^3, 65^3 resolution)
- **LUT Library:**
  - Built-in cinematic LUTs (20+ presets)
  - Film emulation LUTs (Kodak, Fuji, etc.)
  - Custom LUT import/export

    - Community LUT sharing (optional)

LUT Controls:

- LUT intensity/opacity (0-100%)
- LUT blending modes (normal, multiply, screen, overlay)
- LUT preview thumbnails
- Before/after comparison
- LUT favorites/organization

Preset Profiles:

- **Cinematic:** Teal & Orange, Bleach Bypass, Film Noir, Vintage, etc.
- **Film Stock:** Portra 400, Ektar 100, Velvia 50, Tri-X 400, etc.
- **Creative:** High Contrast, Low Contrast, Matte, Faded, Cross-Process
- **Natural:** Neutral, Enhanced, Vivid, Landscape, Portrait

3.5 Detail Enhancement

Priority: P0 (Core Feature)

Sharpening:

- **Amount:** 0-150
- **Radius:** 0.5-3.0 pixels
- **Threshold:** 0-10 (prevents sharpening noise)
- **Masking:** Luminance-based sharpening mask

Clarity & Texture:

- **Clarity:** -100 to +100 (mid-tone contrast)
- **Texture:** -100 to +100 (fine detail enhancement)
- **Dehaze:** -100 to +100 (atmospheric clarity)

Noise Reduction:

- **Luminance NR:** 0-100 (reduces brightness noise)
- **Color NR:** 0-100 (reduces color noise/chroma)
- **Detail Preservation:** 0-100 (maintains edges during NR)
- **Smoothness:** 0-100 (overall smoothing amount)

Grain Addition (Film Simulation):

- **Amount:** 0-100
- **Size:** Fine, Medium, Coarse
- **Roughness:** 0-100 (grain pattern irregularity)

Technical Implementation:

- Multi-scale detail separation (wavelets)
- Edge-aware filtering
- GPU-accelerated convolution kernels
- Real-time preview at reduced resolution
- Full-resolution processing on capture

3.6 Effects & Adjustments

Priority: P1 (Enhanced Feature)

Vignette:

- **Amount:** -100 (lighten) to +100 (darken)
- **Midpoint:** Center position of effect (0-100)
- **Roundness:** -100 (rectangular) to +100 (circular)
- **Feather:** 0-100 (edge softness)
- **Highlights:** Preserve/suppress highlights in darkened areas

Lens Corrections:

- **Distortion:** -100 (barrel) to +100 (pincushion)
- **Vertical Perspective:** -100 to +100
- **Horizontal Perspective:** -100 to +100
- **Rotation:** -45° to +45°
- **Scale:** 50% to 150%

Creative Effects:

- **Tilt-Shift Blur:** Simulate miniature effect
- **Radial Blur:** Motion blur from center point
- **Orton Effect:** Dreamy soft glow
- **Bloom:** Highlight glow/diffusion
- **Glow:** Soft focus effect

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4. COMPOSITION GUIDANCE SYSTEM

4.1 Rule of Thirds

Priority: P0 (Core Feature)

Description: Divides frame into 9 equal parts using 2 horizontal and 2 vertical lines, creating 4 key intersection points for dynamic subject placement.

Features:

- Standard 3×3 grid overlay
- Intersection point indicators (dots, circles, crosses)
- Dynamic line thickness (1-8 pixels)
- Adaptive color with scene-based contrast
- Highlight active intersection nearest to focus point

Customization:

- Line style: Solid, dashed, dotted
- Line color: White, black, yellow, red, cyan, custom RGB
- Opacity: 0-100%
- Intersection point style: None, small dots, large circles, crosses
- Intersection point highlight when focus locked

Mathematical Implementation:

- Precise 1/3 divisions regardless of aspect ratio
- Maintains accuracy during zoom
- Adapts to sensor crop modes
- Scales with digital zoom

4.2 Fibonacci Spiral (Golden Spiral)

Priority: P0 (Core Feature)

Description: Graphical approximation of the golden spiral, constructed using quarter-circle arcs within Fibonacci sequence rectangles (1, 1, 2, 3, 5, 8, 13...), creating natural flow and visual balance.

Features:

- Four orientation options (spiral starts from each corner)
- Dynamic rotation: 0°, 90°, 180°, 270° with smooth animation
- Horizontal/vertical flip capability
- Optional underlying grid structure (Fibonacci rectangles)
- Spiral center point indicator
- Thickness adjustment (1-8 pixels)

Customization:

- Spiral color and opacity
- Show/hide Fibonacci rectangle grid
- Grid line style (solid, dashed)
- Spiral starting corner selection
- Mirror and flip controls
- Animation toggle for educational mode

Mathematical Implementation:

- True Fibonacci proportions: 1:1, 1:2, 2:3, 3:5, 5:8, 8:13
- Golden ratio accuracy ($\varphi \approx 1.618$)
- Aspect ratio adaptation while maintaining proportions
- Sub-pixel rendering for smooth curves

Educational Features:

- Overlay explanation mode (shows number sequence)
- Nature examples (shells, hurricanes, flowers)
- Famous painting/photo analysis with spiral overlay
- Golden ratio calculator

4.3 Golden Ratio Grid

Priority: P1 (Enhanced Feature)

Description: Grid based on golden ratio (1:1.618) divisions rather than simple thirds, creating mathematically harmonious proportions.

Features:

- Phi-based vertical and horizontal divisions
- Primary lines at 1:1.618 ratio from edges
- Secondary lines creating golden rectangles
- Intersection points at ratio positions
- Compatible with all aspect ratios

Implementation:

- Precise φ calculation: (1 + √5) / 2
- Dynamic recalculation for aspect ratio changes
- Multiple golden sections within frame
- Nested golden rectangles option

4.4 Additional Composition Overlays

Priority: P1 (Enhanced Feature)

Diagonal Lines:

- 45° diagonals from all four corners
- "Golden Triangle" composition guide
- Dynamic triangle for leading lines
- Adjustable angle (0-90° in 15° increments)
- Both X patterns and single diagonal options

Center Cross:

- Precise center point marker
- Crosshair with adjustable arms length
- Useful for symmetry and centered compositions
- Magnetic snap when subject centered

Horizon Line:

- Single horizontal guide line
- Adjustable height (0-100% of frame)
- Quick presets: Upper third, center, lower third
- Level indicator (digital bubble level)
- Tilt correction guide

Custom Grid:

- User-defined number of divisions (2-10 horizontal, 2-10 vertical)
- Saved custom grid layouts
- Asymmetric grid options
- Quadrant-based layouts

Dynamic Grid:

- Scene-adaptive grid suggestions
- AI-based composition recommendations
- Leading lines detection and overlay
- Symmetry detection and guide

4.5 Multi-Overlay Support

Priority: P1 (Enhanced Feature)

Capabilities:

- Simultaneous display of up to 3 overlays
- Overlay layering with individual opacity control
- Combine complementary guides (e.g., Rule of Thirds + Diagonal)
- Preset combinations saved as custom layouts
- Quick toggle individual overlays on/off

Overlay Management:

- Layer stack view showing active overlays
- Drag to reorder overlay priority
- Group overlays into presets
- Import/export overlay configurations

4.6 Composition Analysis & Learning

Priority: P1 (Enhanced Feature)

Real-Time Composition Feedback:

- **Balance Meter:** Visual weight distribution indicator
- **Tension Map:** Highlights areas of visual interest/conflict
- **Leading Lines Detection:** AI identifies and highlights natural lines
- **Symmetry Detector:** Shows symmetry axes when present
- **Negative Space Analyzer:** Visualizes subject-to-space ratio

Educational Components:

- **Example Gallery:** 100+ famous photographs analyzed
  - Shows which composition techniques were used
  - Annotations explaining why composition works
  - Includes paintings referenced in client notes
- **Composition Detective Mode:**
  - Upload any photo
  - AI analyzes composition techniques present
  - Overlays applicable grids/spirals
  - Provides composition score and suggestions
- **Daily Composition Challenge:**
  - New prompt each day
  - Specific composition technique to practice
  - Community submissions and feedback (optional)
- **Master Class Series:**
  - Study famous photographers' work
  - Interactive overlays on their images
  - Composition principles explained
  - Practice exercises

Composition Presets Per Genre:

- Portrait: Rule of Thirds with center emphasis
- Landscape: Golden Ratio with horizon placement
- Architecture: Golden Ratio + vertical/horizontal guides
- Street: Dynamic diagonals with rule of thirds
- Macro: Fibonacci spiral emphasis
- Sports: Diagonal action lines

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5. ADVANCED CAMERA FEATURES

5.1 HDR & Dynamic Range Control

Priority: P0 (Core Feature)

HDR Processing:

- **Real-Time HDR Preview:** Live preview of merged exposure
- **Multi-Frame HDR:** 3, 5, or 7 shot automatic bracketing
- **Exposure Fusion:** Smart blending algorithm
- **Tone Mapping:** Multiple algorithms (Natural, Artistic, Dramatic)
- **Deghost:** Automatic moving object handling

Dynamic Range Tools:

- **Highlight Protection:** Preserves detail in bright areas
- **Shadow Recovery:** Extracts detail from dark areas
- **Local Tone Mapping:** Area-specific exposure adjustment
- **HDR Strength Control:** 0-100% effect intensity

User Controls:

- HDR mode: Off, Auto, Manual bracketing
- Bracket spacing: 1, 2, 3 EV
- Alignment: Auto-align bracketed shots
- Output: Single HDR JPEG, DNG stack, or both

5.2 Long Exposure & Motion Effects

Priority: P1 (Enhanced Feature)

Computational Long Exposure:

- **Duration:** 1 second to 5 minutes
- **Implementation:** Multi-frame averaging with motion detection
- **Effects:**
  - Light trails (vehicle lights, stars)
  - Water smoothing (waterfalls, oceans)
  - Cloud movement blur
  - Crowd removal (moving people disappear)

Motion Blur Synthesis:

- Simulated shutter drag effect
- Directional motion blur for selected subjects

- Speed-based blur intensity
- Motion vector visualization

ND Filter Simulation:

- Equivalent ND strength: ND2 to ND1000
- Reciprocity calculator (adjusts shutter/aperture)
- Real-time brightness preview
- Optional graduated ND effect

5.3 Focus Stacking

Priority: P1 (Enhanced Feature)

Automatic Focus Stacking:

- Set near and far focus points
- Automatically captures multiple focus planes (5-50 shots)
- Real-time focus distance visualization
- Live preview of stacked result

Manual Focus Stacking:

- Manual focus control between shots
- Focus peaking guide for next shot
- Alignment markers
- Shot progress indicator

Export Options:

- Merged all-in-focus image
- Individual focus planes
- DNG stack for external processing
- Depth map extraction

5.4 Time-Lapse & Interval Timer

Priority: P1 (Enhanced Feature)

Interval Photography:

- Interval range: 1 second to 24 hours
- Duration: Unlimited (constrained by storage/battery)
- Frame count calculator
- Playback preview at various speeds

Time-Lapse Features:

- Deflicker algorithm (consistent exposure across frames)
- Holy grail time-lapse (day-to-night transitions)
- Motion tracking time-lapse (subject stays centered)
- Exposure ramping for lighting changes

Export:

- 4K video at 24/30/60fps
- Individual frames as JPEG/DNG
- ProRes/H.265 codec options

5.5 Astrophotography Mode

Priority: P2 (Future Enhancement)

Star Photography:

- Extended long exposure (30s-5min)
- Star trail stacking
- Star tracking (compensates for Earth rotation)
- Hot pixel removal
- Dark frame subtraction

Moon/Planet Photography:

- Telephoto lens optimization
- Multi-frame lucky imaging
- Surface detail enhancement
- Video-to-still stacking

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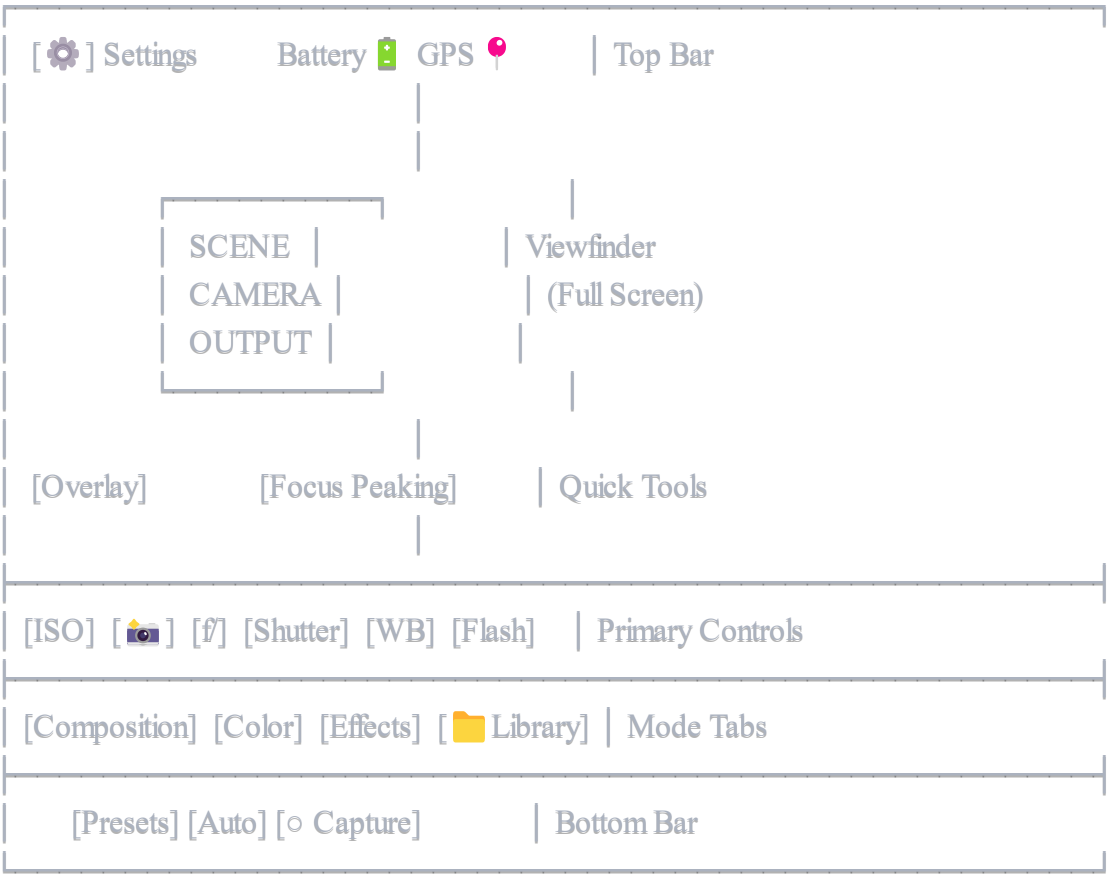
6. USER INTERFACE & EXPERIENCE

6.1 Camera View Interface Design

Philosophy: Minimal, professional, quick-access controls that don't obstruct the scene while providing immediate access to all critical functions.

Screen Layout:





Control Zones:

Top Bar (Persistent):

- Settings access (gear icon)
- Current lens indicator (0.5x, 1x, 2x, 3x)
- Battery and storage indicators
- GPS/location toggle
- Level indicator (digital bubble)

Viewfinder Area (Full Screen):

- Clean viewfinder with minimal obstruction
- Tap to focus
- Pinch to zoom
- Swipe gestures for quick adjustments
- Composition overlays (when enabled)
- Exposure tools (histogram, zebras, waveform - toggleable)

Quick Access Tools (Floating/Collapsible):

- Focus peaking toggle
- Overlay selector
- Grid toggle
- Histogram toggle
- Zebra patterns toggle
- Level/horizon toggle

Primary Control Strip:

- ISO dial/button
- Shutter speed dial/button
- Aperture (f-stop) dial/button
- White balance button
- Flash/torch toggle
- Exposure compensation dial

Mode Tabs:

- Composition (overlays and guides)
- Color (grading and LUTs)
- Effects (lens effects, filters)
- Library (recent captures)

Interaction Patterns:

- **Tap:** Open full control panel
- **Long Press:** Quick adjustment mode (drag to change value)
- **Double Tap:** Reset to auto/default
- **Swipe:** Quick value adjustment without opening panel

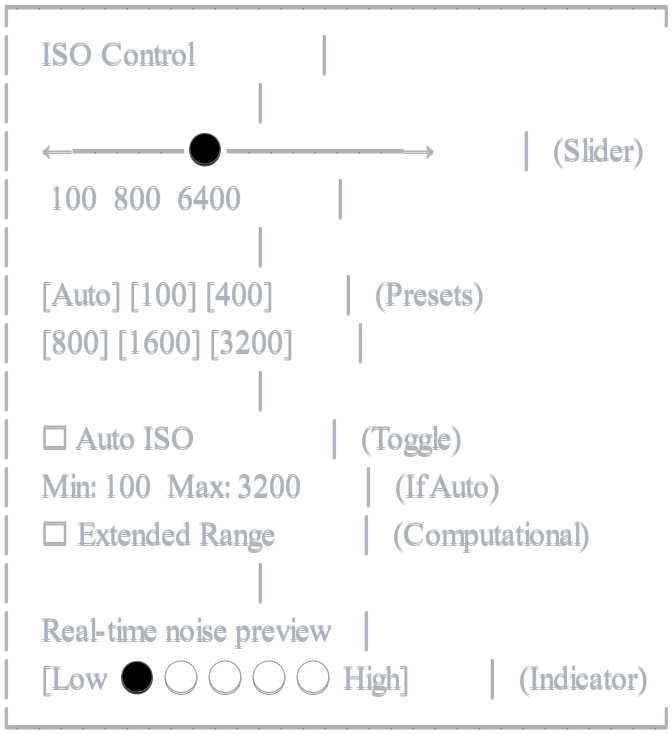
6.2 Advanced Control Panels

Expandable Control Interfaces:

Each primary control opens a detailed panel when tapped:

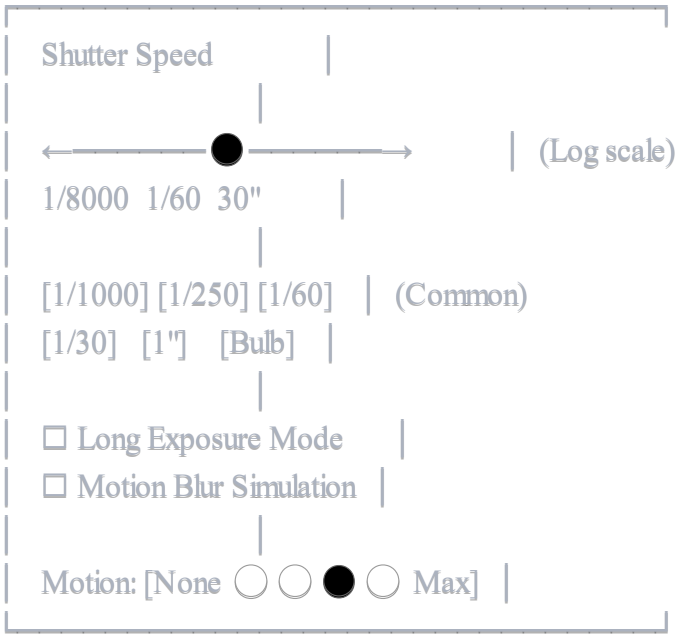
ISO Panel:





Shutter Speed Panel:





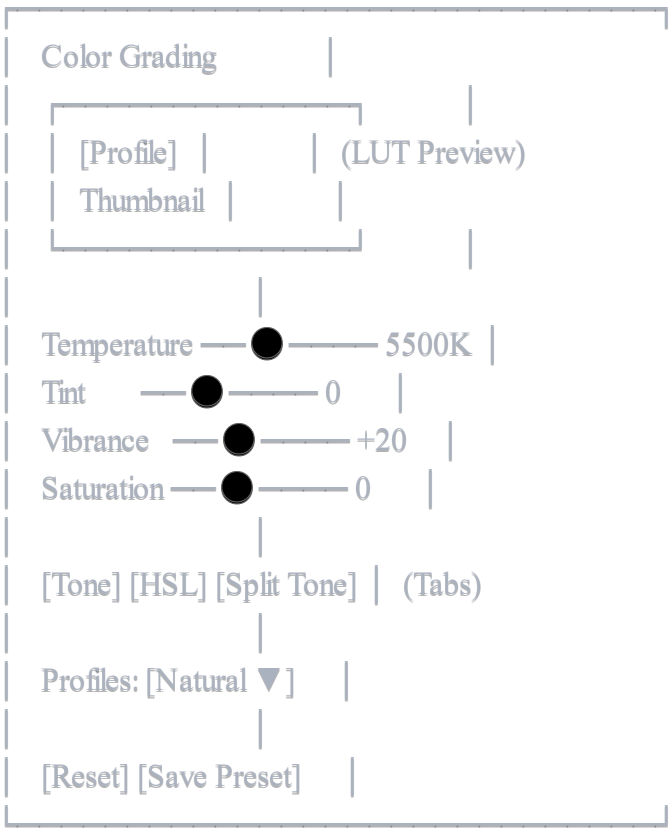
Composition Panel:





Color Grading Panel:





6.3 Gesture Controls

Global Gestures (Viewfinder):

- **Single Tap:** Set focus point
- **Double Tap:** Toggle all overlays on/off
- **Long Press:** Lock exposure and focus (AE-L/AF-L)
- **Two-Finger Tap:** Quick settings menu
- **Pinch:** Zoom in/out (optical + digital)
- **Two-Finger Rotate:** Rotate active overlay (where applicable)
- **Three-Finger Swipe Up:** Show/hide all UI
- **Three-Finger Swipe Down:** Quick access exposure tools

Control-Specific Gestures:

- **Swipe on ISO/Shutter/Aperture buttons:** Quick adjust without opening panel
- **Swipe on Histogram:** Adjust exposure compensation
- **Swipe on Overlay:** Adjust opacity
- **Drag on Viewfinder (when overlay visible):** Reposition overlay (if applicable)

Hardware Button Integration:

- **Volume Up:** Toggle overlay visibility
- **Volume Down:** Alternative shutter button
- **Volume Up (Long Press):** Start/stop video recording

- **Volume Up + Down:** Quick HDR capture

6.4 Presets & Workflow Management

Preset System:

What Can Be Saved:

- All camera settings (ISO, shutter, aperture, WB)
- Composition overlays (type, color, position)
- Color grading (LUT, adjustments, tone curves)
- Lens effects (aperture, bokeh, distortion)
- All image processing settings

Preset Management:

- Unlimited custom presets
- Preset naming and organization
- Preset categories (Portrait, Landscape, Street, etc.)
- Preset thumbnails (sample image)
- Import/export presets as files
- Cloud sync presets (optional)
- Community preset sharing (optional)

Quick Preset Access:

- Favorite bar with 5 quick-access slots
- Preset selector drawer (swipe up)
- Search presets by name or tag
- Recent presets history
- Auto-save last used settings

Preset Application:

- Instant load (apply all settings)
- Selective load (choose which settings to apply)
- Blend between presets (morph settings)
- A/B comparison (switch between two presets)

6.5 Export & Workflow Options

Capture Modes:

- Single shot
- Burst mode (up to 20fps)
- Bracketing (exposure, focus, white balance)
- RAW + JPEG pairs
- RAW only
- JPEG only (with embedded adjustment data)

Export Formats:

- **JPEG:** Quality slider (60-100%), size selection
- **PNG:** Lossless compression

- **HEIC:** Modern efficient format (iOS)
- **DNG/RAW:** With all adjustment metadata
- **TIFF:** 16-bit uncompressed (for maximum quality)

Export Destinations:

- Device photo library
- App internal gallery
- Direct share to social media (Instagram, etc.)
- Send to editing apps (Lightroom, Snapseed, etc.)
- Cloud storage (iCloud, Google Drive, Dropbox)
- AirDrop/Share (iOS) / Nearby Share (Android)

Batch Operations:

- Apply same preset to multiple photos
- Batch export with consistent settings
- Batch format conversion
- Batch resize for social media

6.6 Professional Display Tools

Exposure Analysis Tools:

- **Live Histogram:** RGB + Luminance, position adjustable
- **Zebra Patterns:** Configurable threshold (95-105% for highlights, 0-10% for shadows)
- **False Color:** IRE/stop zones with color coding
- **Waveform Monitor:** Luminance and RGB parade modes
- **Vectorscope:** Color saturation and hue distribution

Focus Tools:

- **Focus Peaking:** Adjustable colors (red, yellow, green, white, blue)
- **Focus Magnifier:** Pinch to zoom 2x, 5x, 10x on focus point
- **Split Screen:** Dual view with different magnifications
- **Focus Distance Scale:** Meters/feet with depth of field indicator

Composition Tools:

- **Safe Areas:** Show broadcast safe zones (action/title safe)
- **Aspect Ratio Masks:** Preview different crop ratios
- **Grid Overlay Combination:** Multiple overlays simultaneously
- **Horizon Level:** Real-time tilt indicator with degree measurement

These tools can be individually toggled and repositioned on screen.

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7. TECHNICAL ARCHITECTURE & OPTIMIZATION

7.1 Performance Requirements

Frame Rate Targets:

- **Preview (1080p):** Minimum 30fps, target 60fps sustained

- **Preview (4K):** Minimum 24fps, target 30fps
- **Capture Processing:** Full-resolution image ready within 3 seconds
- **Effect Adjustments:** Changes reflected within 100ms
- **App Launch:** Camera-ready in under 2 seconds cold start

Device Tier Strategy:

Tier 1: Flagship Devices (2023+)

- Examples: iPhone 15 Pro, Pixel 8 Pro, Galaxy S24 Ultra
- Capabilities: All features at maximum quality
- Target: 60fps preview with all effects active
- Multi-frame processing: Real-time HDR, focus stacking, super-resolution

Tier 2: Premium Devices (2021-2023)

- Examples: iPhone 13 Pro, Pixel 6 Pro, Galaxy S21
- Capabilities: All features with quality/speed balance
- Target: 30-60fps adaptive framerate
- Multi-frame processing: Available but slower

Tier 3: Mid-Range & Older Flagships (2019-2021)

- Examples: iPhone 11, Pixel 5, Galaxy S20
- Capabilities: Core features with performance optimizations
- Target: 30fps with adaptive quality reduction
- Some advanced features limited or preview-only

Tier 4: Minimum Spec Devices

- Minimum: iPhone XR/Android equivalent (2018+)
- Capabilities: Essential features only
- Target: 24-30fps, simplified effects
- Advanced computational features disabled or capture-only

Adaptive Performance System:

- Automatic device capability detection on first launch
- Dynamic feature enablement based on available resources
- Real-time performance monitoring and adjustment
- Battery-aware processing (reduce intensity below 20% battery)
- Thermal monitoring (throttle processing if device overheating)
- Memory management (reduce cache size if memory constrained)

7.2 Rendering Pipeline Optimization

Multi-Threaded Architecture:



[Camera Input Thread]
↓
[Depth Processing Thread] → [Depth Map Cache]
↓
[Image Processing Thread Pool] (4-8 threads)
↓
[GPU Rendering Thread] → [Frame Buffer]
↓
[Display Thread] → Screen Output

GPU Acceleration:

- **iOS:** Metal framework for compute shaders
- **Android:** Vulkan (primary) with OpenGL ES fallback
- **Shader Pipeline:**
  - Depth map generation shaders
  - Bokeh rendering shaders
  - Color grading LUT shaders
  - Tone mapping shaders
  - Overlay compositing shaders

Caching Strategy:

- **Depth Maps:** Cache for 5 most recent frames, predict next frame depth
- **LUT Textures:** Preload and cache all active LUTs (GPU memory)
- **Processed Overlays:** Rasterize overlays once, composite each frame
- **Settings State:** Immediate memory for all adjustments
- **Preview Frames:** Triple buffering for smooth display

Lazy Computation:

- Only process visible effects
- Defer full-resolution processing until capture
- Use lower resolution for real-time preview where imperceptible
- Progressive rendering: Show quick preview, refine in background

7.3 Camera API Integration

iOS (AVFoundation):

- `AVCaptureSession` for camera management
- `AVCaptureDevice` for lens and sensor control
- `AVCapturePhotoOutput` for still capture with RAW support
- `AVCaptureVideoDataOutput` for frame-by-frame processing
- `AVDepthData` for hardware depth map access
- Manual controls via `AVCaptureDevice.exposureMode`, `focusMode`, etc.

Android (CameraX + Camera2):

- `CameraX` for simplified camera lifecycle management
- `Camera2 API` for advanced manual controls

- `ImageAnalysis` use case for real-time frame processing
- `ImageCapture` use case for high-quality still capture
- `DepthOutput` for depth sensor access (where available)
- Manual control via `CaptureRequest` with `CONTROL_MODE_OFF`

Cross-Platform Abstraction:

- Unified camera interface layer
- Platform-specific implementations
- Capability detection and graceful fallback
- Consistent behavior across platforms where possible

7.4 Storage & Data Management

Local Storage:

- **App Size:** Target under 100MB base, 200MB with assets
- **Cache Management:** Automatic cleanup of old previews and temp files
- **User Data:** Settings, presets, favorites stored in SQLite database
- **Image Library:** Internal gallery with metadata, linked to system photos

File Formats:

- **JPEG:** Standard compressed output
- **HEIC:** Efficient iOS format with metadata
- **DNG:** Adobe RAW format with sidecar for adjustments
- **XMP:** Sidecar files for non-destructive editing data

Cloud Integration (Optional):

- Preset sync across devices
- Settings backup and restore
- Community preset sharing
- Photo backup (partner with existing services)

7.5 Battery & Thermal Management

Power Optimization:

- **Display:** OLED-optimized dark UI to reduce power
- **Camera:** Release camera resources when app backgrounded
- **Processing:** Reduce frame rate when no user interaction
- **Background:** Minimal background activity
- **Battery Levels:**
    - 50%: Full performance mode
    - 20-50%: Balanced mode (60fps → 30fps, reduce some effects)
    - <20%: Power saving mode (30fps, essential features only)

Thermal Management:

- Monitor device temperature via system APIs
- Reduce processing intensity if temperature critical
- Warn user if extended heavy use detected

- Suggest breaks for device cooling
- Prioritize capture quality over preview quality when hot

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8. QUALITY ASSURANCE & EDGE CASES

8.1 Device-Specific Challenges

Multi-Camera Systems:

- Handle seamless switching between lenses (0.5x, 1x, 2x, 3x, 5x)
- Maintain overlay position during lens switch
- Adjust depth map strategy per lens
- Compensate for different lens characteristics

Depth Sensor Variations:

- **Dual Camera:** Stereo depth calculation
- **LiDAR:** Direct depth measurement (iPhone Pro, some Androids)
- **Single Camera:** Monocular AI depth estimation
- **No Depth Hardware:** Disable depth-dependent features gracefully

Screen Types:

- **OLED:** High contrast, true blacks (optimize UI for power)
- **LCD:** Standard calibration
- **High Refresh Rate:** 120Hz/144Hz smooth UI animations
- **Notch/Punch-hole:** UI layout adaptation

8.2 Edge Case Handling

Low Light Scenarios:

- Auto-adjust overlay contrast for visibility
- Boost preview brightness without affecting capture
- Extended ISO range warnings
- Noise preview in real-time
- Suggest tripod/stabilization for long exposure

High Contrast Scenes:

- HDR preview and capture recommendations
- Highlight/shadow recovery suggestions
- Exposure compensation guidance
- Zone metering assistance

Fast-Moving Subjects:

- Continuous AF tracking
- Burst mode for action
- Shutter speed recommendations
- Motion blur warnings

Extreme Focal Lengths:

- Digital zoom quality warnings
- Stabilization recommendations
- Composition guide scaling
- Resolution/crop indicators

Orientation Changes:

- Smooth overlay rotation animation
- Setting preservation across orientations
- UI element repositioning
- No jarring transitions

8.3 Error Handling & User Feedback

Camera Access Issues:

- Clear permission request explanations
- Graceful degradation if permissions denied
- Settings deep-link for permission changes

Hardware Limitations:

- Feature unavailability explained clearly
- Suggest device requirements for unavailable features
- Never crash due to missing hardware
- Offer software alternatives where possible

Performance Issues:

- Automatic quality reduction with user notification
- "Performance Mode" toggle for manual control
- Clear indicators when features temporarily disabled
- Recovery when resources available again

Storage Issues:

- Storage space warnings before capture
- Automatic cache cleanup
- Option to save to external storage (Android)
- Clear storage management tools

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9. PLATFORM-SPECIFIC CONSIDERATIONS

9.1 iOS-Specific Features

Native Integration:

- iPhone camera control button (iPhone 15 Pro)
- Live Photos compatibility
- ProRAW and ProRes support
- Cinematic Mode integration (video)
- Portrait Lighting effects integration
- Focus and Depth Control API

- Smart HDR 5 integration
- Photonic Engine leverage

iOS UI/UX:

- Dynamic Island integration (notifications, timers)
- iOS design language and patterns
- SF Symbols for iconography
- Haptic Engine feedback
- Widgets for quick capture/presets
- Shortcuts app integration
- Handoff between devices

App Store Optimization:

- Screenshots showcasing iOS-specific features
- Privacy nutrition labels
- App Clips for quick try
- TestFlight beta program

9.2 Android-Specific Features

Native Integration:

- Google Photos integration
- Night Sight integration (Pixel)
- Camera2 API advanced controls
- Display P3 color space support (newer devices)
- Material You theming
- Quick Settings tile
- Foldable device support (Galaxy Z Fold, Flip)

Android UI/UX:

- Material Design 3 guidelines
- Dynamic color theming
- Edge-to-edge display
- Gesture navigation support
- Split-screen multitasking
- Adaptive layouts for tablets/foldables

Fragmentation Management:

- Minimum API level 29 (Android 10)
- Graceful feature detection for various OEMs
- Samsung OneUI optimizations
- Pixel-specific enhancements
- Extensive device testing matrix

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10. SUCCESS METRICS & ANALYTICS

10.1 Key Performance Indicators

Technical Performance:

- Average frame rate across device tiers
- App crash rate (<0.1% sessions)
- Average load time (<2s)
- Battery drain per 30-min session (<15%)
- Memory usage (within OS guidelines)

User Engagement:

- Daily Active Users (DAU)
- Session duration (target: 10-15 minutes)
- Photos captured per session (target: 15-25)
- Feature usage rates (which overlays, effects, controls used most)
- Preset creation and reuse rates
- Return usage frequency

Quality Metrics:

- Photo resolution distribution
- RAW vs JPEG capture ratio
- Effect intensity distributions (are users using effects?)
- Export destinations (where do photos go?)
- Time spent adjusting vs capturing

Learning & Education:

- Tutorial completion rates
- Composition guide usage patterns
- Example gallery views
- Preset exploration behavior

10.2 Privacy-First Analytics

Data Collection Principles:

- No personally identifiable information
- No image data sent to servers
- Aggregated, anonymized usage statistics only
- User opt-in for analytics
- Clear privacy policy

What We Track:

- Feature usage counts (anonymous)
- Performance metrics (crash reports)
- Device capability distribution
- Popular presets (anonymous ID only)
- Session duration and frequency

What We Never Track:

- Actual photos or image content
- Location data (unless explicitly enabled by user)
- User identity or contact information
- Social connections or sharing patterns

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11. MONETIZATION (OPTIONAL FUTURE CONSIDERATION)

While the core app can launch as a premium paid application or free with in-app purchases, consider these models:

Option 1: Freemium Model

Free Tier:

- Full manual camera controls
- Rule of Thirds and basic overlays
- 3 preset slots
- JPEG export
- Basic color adjustments

Pro Tier ($9.99 one-time or $3.99/month):

- All composition overlays (Fibonacci, Golden Ratio, custom)
- Unlimited presets with cloud sync
- Advanced lens effects (bokeh, focal length simulation)
- RAW/DNG capture and export
- All LUTs and film profiles
- Advanced color grading (HSL, curves, split-toning)
- HDR and long exposure modes
- Priority support and early access to features

Option 2: Premium App Model

- One-time purchase $14.99-19.99
- All features included
- No subscriptions
- Lifetime updates

Option 3: Professional Edition

- Standard App: $9.99 (for enthusiasts)
- Pro Edition: $29.99 (for professionals)
    - Includes advanced features
    - Desktop companion app for preset management
    - Direct integration with professional workflows
    - Commercial use license

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12. DEVELOPMENT ROADMAP

Phase 1: Core Foundation (Months 1-5)

Goal: Functional camera app with manual controls and real-time preview

Deliverables:

- ✅ Camera API integration (iOS & Android)
- ✅ Manual controls (ISO, shutter, aperture simulation, WB, focus)
- ✅ Real-time processing pipeline (basic)
- ✅ Composition overlays (Rule of Thirds, Fibonacci Spiral)
- ✅ Basic exposure tools (histogram, zebras)
- ✅ JPEG capture and export
- ✅ Preset system (save/load camera settings)
- ✅ Core UI/UX implementation
- ✅ Performance optimization for flagship devices

Testing:

- iPhone 15 Pro, 14 Pro
- Pixel 8 Pro, Galaxy S24 Ultra
- Basic performance benchmarks

Phase 2: Computational Photography (Months 6-9)

Goal: Advanced lens simulation and image processing

Deliverables:

- ✅ Depth map generation (multi-method)
- ✅ Bokeh rendering engine
- ✅ Focal length emulation with super-resolution
- ✅ Real-time color grading engine
- ✅ LUT system and preset library
- ✅ Tone curves and advanced adjustments
- ✅ HSL color controls
- ✅ RAW/DNG capture
- ✅ HDR capture and processing
- ✅ Lens effects (distortion, vignette, CA)

Testing:

- Expand device testing to Tier 2 devices
- Performance optimization for mid-range
- Quality validation vs. professional cameras

Phase 3: Advanced Features & Polish (Months 10-12)

Goal: Professional-grade features and workflow tools

Deliverables:

- ✅ Focus stacking
- ✅ Long exposure and ND filter simulation
- ✅ Time-lapse and interval timer
- ✅ Advanced composition analysis tools
- ✅ Educational content and example gallery
- ✅ Gesture controls and shortcuts
- ✅ Batch processing and export
- ✅ Cloud preset sync
- ✅ Additional overlays (Golden Ratio, diagonals, custom grids)
- ✅ Video recording with overlays (if scope allows)

Testing:

- Comprehensive device testing (Tiers 1-4)
- Real-world usage with photographers
- Performance and battery optimization
- User feedback integration

Phase 4: Refinement & Launch Prep (Months 13-15)

Goal: Production-ready app with excellent stability

Deliverables:

- ✅ Bug fixes and stability improvements
- ✅ Performance optimization across all device tiers
- ✅ Localization (5+ languages)
- ✅ Accessibility features (VoiceOver, TalkBack, etc.)
- ✅ Onboarding and tutorials
- ✅ Marketing materials and app store assets
- ✅ Beta testing program (TestFlight/Play Store Beta)
- ✅ Documentation and help center
- ✅ Final QA and certification

Launch:

- iOS App Store submission
- Google Play Store submission
- Press and influencer outreach
- Community building

Post-Launch: Continuous Improvement

Ongoing:

- User feedback integration
- Bug fixes and performance updates
- New preset and LUT packs
- Additional composition guides
- Community features (optional)
- Desktop companion app (future)
- Advanced AI features (scene recognition, auto-composition)

13. TECHNICAL DEPENDENCIES & THIRD-PARTY LIBRARIES

Core Dependencies

iOS:

- AVFoundation (camera, audio, media)
- CoreImage (image processing filters)
- Metal (GPU acceleration)
- Vision (AI depth estimation, face detection)
- CoreML (machine learning models)
- Photos/PhotoKit (library access)

Android:

- CameraX & Camera2 API
- RenderScript/Vulkan (GPU processing)
- ML Kit (depth, face detection)
- TensorFlow Lite (custom ML models)
- Jetpack Compose (modern UI)

Cross-Platform (if using React Native/Flutter):

- React Native Camera or Flutter Camera plugin
- Custom native modules for advanced features
- Bridge to native processing pipelines

Third-Party Considerations

Image Processing:

- Custom shaders for real-time effects
- OpenCV (optional, for advanced computer vision)
- FFmpeg (if video processing needed)

Storage & Sync:

- Realm or SQLite for local database
- Firebase or AWS for cloud features (optional)

Analytics:

- Privacy-focused analytics (Plausible, Mixpanel with anonymization)
- Crashlytics for crash reporting

Avoid:

- Heavy ML frameworks (keep app size small)
- Excessive third-party dependencies (reduce attack surface)
- Analytics that compromise privacy

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14. OPEN QUESTIONS & DECISIONS NEEDED

Critical Decisions

1. **Platform Priority:** Launch iOS first, Android first, or simultaneous?
    - Recommendation: iOS first (easier camera API, smaller device fragmentation)
2. **Technology Stack:** Native (Swift/Kotlin) or Cross-Platform (React Native/Flutter)?
    - Recommendation: Native for best performance and camera control
3. **Monetization:** Free, Freemium, Premium paid, or Subscription?
    - Recommendation: Premium paid ($14.99) or Freemium with Pro upgrade
4. **Target Audience Positioning:** Consumer/enthusiast or professional photographer?
    - Recommendation: Enthusiast focus with professional capabilities
5. **Feature Scope for MVP:** Include all Phase 1+2, or more minimal launch?
    - Recommendation: Phase 1 + core Phase 2 (depth/bokeh) for strong differentiation
6. **Brand Identity:** App name, icon, visual style?
    - Consider: Professional, clean, photography-focused aesthetic
7. **Community Features:** Include sharing/community from start or add later?
    - Recommendation: Add later, focus on core camera experience first
8. **Educational Content:** How much to include at launch?
    - Recommendation: Essential guides + example gallery, expand post-launch

Technical Validation Needed

- Confirm achievable frame rates on target devices
- Validate depth map quality across devices
- Test bokeh rendering performance
- Verify real-time color grading speed
- Assess battery impact of continuous use
- Confirm storage requirements for typical session

APPENDIX

A. Composition Techniques - Complete Reference

Rule of Thirds: Mathematical division of frame into 9 equal rectangles. Placing subjects at intersections or along lines creates tension and interest versus centered composition.

Fibonacci Spiral (Golden Spiral): Based on Fibonacci sequence (1, 1, 2, 3, 5, 8, 13, 21...). Each number is sum of previous two. Creates rectangles with ratios approaching golden ratio ($\varphi \approx 1.618$). Quarter-circles in each rectangle form spiral found in nature (shells, hurricanes, flower petals, galaxy arms). Guides viewer's eye naturally through image.

Golden Ratio: Mathematical ratio where $(a+b)/a = a/b \approx 1.618$. Creates harmonious proportions considered aesthetically pleasing. Found in nature, architecture, classical art. Divides frame at approximately 61.8% from either edge.

Leading Lines: Natural or man-made lines that draw viewer's eye to subject or through composition. Can be roads, rivers, fences, architectural elements. Diagonal lines create more dynamic tension than horizontal/vertical.

Symmetry & Patterns: Bilateral symmetry creates balance and calm. Breaking symmetry creates interest. Repeating patterns with intentional break draws attention.

Negative Space: Empty space around subject. Proper balance of positive (subject) and negative space creates emphasis and breathing room.

Framing: Using natural frames (doorways, windows, arches, branches) to draw attention to subject and add depth.

Perspective & Depth: Foreground, middle ground, background elements create three-dimensional feel. Leading lines to vanishing point enhance depth perception.

Reference Resources:

- Research paintings from masters (as client notes suggest 8-13 examples)
- Ansel Adams' landscape compositions
- Henri Cartier-Bresson's decisive moment
- Steve McCurry's use of golden ratio
- Study nature: Nautilus shells, sunflowers, hurricanes (Fibonacci examples)

B. Competitive Landscape Analysis

Professional Camera Apps:

- **Halide:** $50/year, excellent manual controls, limited computational features
- **ProCamera:** $15 one-time, good controls, outdated interface
- **Moment Pro:** $6/month, solid features, requires Moment lenses for best results
- **Camera+ 2:** $8 one-time, good but aging platform

Computational Photography:

- **Lightroom Mobile:** Free/subscription, excellent editing but not camera-focused
- **Snapseed:** Free, great editing, no camera controls
- **VSCO:** $30/year, film presets focus, limited camera features

Composition Tools:

- Very limited market
- Most cameras have basic grid only
- No real-time lens simulation + composition + grading combo

Our Differentiation:

- Complete professional system in one app
- Real-time everything (lens effects, grading, composition)
- Educational focus (learning while shooting)
- No need for external editing
- Better than built-in camera + separate editing workflow

C. Glossary of Photography Terms

For reference when communicating with photographers:

- **Aperture (f-stop):** Opening in lens controlling light and depth of field
- **Bokeh:** Quality of out-of-focus areas
- **Circle of Confusion:** Blur circle size determining sharpness
- **Depth of Field (DOF):** Distance range in acceptable focus
- **Dynamic Range:** Ratio between darkest and brightest tones
- **Exposure Triangle:** ISO, shutter speed, aperture relationship
- **Golden Hour:** Hour after sunrise/before sunset (warm light)
- **HDR:** High Dynamic Range, merging multiple exposures

- **Histogram:** Graphical representation of tonal distribution
- **Hyperfocal Distance:** Focus distance maximizing DOF
- **IRE:** Video brightness scale (0-100+)
- **LUT:** Look-Up Table for color grading
- **Reciprocity:** Equivalent exposures with different settings
- **Stop:** One stop = doubling or halving light
- **Tone Mapping:** Compressing HDR to displayable range
- **Waveform:** Brightness distribution across frame width
- **Zebra Pattern:** Overexposure warning overlay

FINAL SUMMARY

This application represents a comprehensive solution to the client's vision: a single mobile app that eliminates the need to carry professional camera equipment while providing complete creative control over every aspect of photography.

Core Pillars:

1. **Lens Simulation:** Professional depth of field, bokeh, and focal length emulation
2. **Manual Camera Controls:** Full exposure triangle control with professional tools
3. **Real-Time Processing:** What-you-see-is-what-you-get color grading and effects
4. **Composition Guidance:** Educational overlays with client's requested Rule of Thirds and Fibonacci Spiral
5. **Professional Workflow:** Presets, batch processing, RAW support

Technical Feasibility: Achievable with modern mobile hardware and computational photography techniques, with adaptive performance across device tiers.

Market Position: Unique combination of features not currently available in single app.

User Experience: Intuitive interface balancing professional power with accessibility.

This PRD provides complete technical specification for development while maintaining flexibility for refinement based on testing and user feedback.